# Division Decoded

The mission where the leftover needs a decision.

THE BIG QUESTION

## When you have a remainder, what do you do with it — and who decides?

HANDS-ON PROJECT

### Snack Run

Split real quantities among real people. Defend every remainder decision out loud.

FLUENCY GAME

### Remainder Race

Deal cards, divide, score the leftover. Biggest remainder takes the round.

AXIOM · MISSION COMMANDER, NINE TOURS

"Brock — the leftover is not a mistake. It's information. Most people never learn to read it."

NAME ______________

DATE ______________

# Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

**This is not a test.** Nothing here counts for a grade. It exists to delete work you don't need. If you don't know something, write “not yet” and move on — that's a useful answer.

STRAND A
FACTS

A1 $42 \div 6 =$ ..............................

A2 $56 \div 8 =$ ..............................

STRAND B
MISSING SIDE

B1 Draw $84 \div 4$ as a rectangle.

B2 $156 \div 6 =$ ________

STRAND C
REMAINDERS

C1 $94 \div 4 =$ _____ r _____

C2 26 kids, vans hold 6. How many vans? Why?

STRAND D
LONG DIVISION

D1 $372 \div 3 =$ _______

D2 $845 \div 4 =$ _____ r _____

STRAND E
DIVISIBILITY

E1 Which of 2, 3, 5, 9 divide 234?

E2 Does 7 divide 91? How do you know?

STRAND F
REASONING

F1 100 stickers, 7 kids. What happens to the extras?

F2 $\square 6 \div 4 = 24$ $\square =$ _____

# Skip Chart

Score the pre-assessment, then cross weeks off the calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

| STRAND | ITEMS | COVERS | SCORE | IF 2/2, REMOVE |
|---|---|---|---|---|
| **A · Facts** | A1<br>A2 | Division facts to 81 | ___ / 2 | Warm-Up tier drops to 3 items all unit |
| **B · Missing side** | B1<br>B2 | Week 1 · lessons 1.1–1.2 | ___ / 2 | Days 1.1 and 1.2 — keep Friday enrichment |
| **C · Remainders** | C1<br>C2 | Week 1 · 1.3 and Week 2 · 2.1–2.3 | ___ / 2 | Days 2.1–2.3. Never skip 2.4 — remainder-as-answer is new to almost everyone. |
| **D · Long division** | D1<br>D2 | Week 1 · 1.4 and Week 3 | ___ / 2 | Week 3 days 3.1–3.3. Keep 3.4, the multiply-back check. |
| **E · Divisibility** | E1<br>E2 | Week 4 · lessons 4.1–4.2 | ___ / 2 | Days 4.1–4.2. Never skip 4.4 (two-digit divisors). |
| **F · Reasoning** | F1<br>F2 | Week 2 · 2.1–2.4 and Week 5 · 5.1 | ___ / 2 | Nothing. Raise the Challenge tier instead. |

## Where the saved time goes

Never back into the calendar. Compacted time is spent on the Challenge tier, the Snack Run project, or an early start on Mission 03. Write the reallocation here so it doesn't quietly evaporate:

............................................................

............................................................

## Revised Mission 02 plan

WEEK 1 ☐ full ☐ compressed ☐ skip

WEEK 2 ☐ full ☐ compressed ☐ skip

WEEK 3 ☐ full ☐ compressed ☐ skip

WEEK 4 ☐ full ☐ compressed ☐ skip

WEEK 5 ☐ full ☐ compressed ☐ skip

NAME ___________

DATE ___________

# Two Kinds of Division

Deal them into rows, or deal them into groups. Same picture, two questions.

## Warm-Up

6 ITEMS · 5 MINUTES · FACTS YOU ALREADY OWN

| | | | | | |
|---|---|---|---|---|---|
| 42 ÷ 6 | 56 ÷ 8 | 63 ÷ 9 | 36 ÷ 4 | 72 ÷ 8 | 45 ÷ 5 |
| ______ | ______ | ______ | ______ | ______ | ______ |

42 ÷ 6 **Sharing:** 6 equal rows — how many in a row? **Grouping:** groups of 6 — how many groups? Both: 7

## Core

DRAW THE RECTANGLE. SPLIT THE AREA INTO FRIENDLY PIECES.

1. 84 ÷ 4 = ______

2. 96 ÷ 6 = ______

3. 75 ÷ 5 = ______

4. 90 ÷ 6 = ______

5. 78 ÷ 3 = ______

For number 5, write the two pieces you split 78 into:

................................................

## Challenge

NOT GRADED

**C1.** A rectangle has area 96 and one side 6. How long is the other side? Show how you found it.

................................................

**C2.** 72 pencils go into boxes of 8. How many boxes? Is that a sharing question or a grouping question?

................................................

**C3.** Write two stories for 48 ÷ 8 — one sharing, one grouping. They should sound completely different.

................................................

**AXIOM:** Division is the same rectangle read from a different corner. Say the multiplication fact out loud first.

NAME ___________

DATE ___________

# The Missing Side

156 ÷ 6 is hard. Twenty sixes and six sixes are easy. That's the whole trick.

## ✦ Warm-Up

| | | | | | |
|---|---|---|---|---|---|
| 24 ÷ 3 | 54 ÷ 6 | 81 ÷ 9 | 32 ÷ 8 | 49 ÷ 7 | 60 ÷ 5 |
| ____ | ____ | ____ | ____ | ____ | ____ |

WORKED EXAMPLE

156 ÷ 6

| 20 sixes 120 | 6 sixes 36 |
|---|---|

20 + 6 = 26

Add the *widths*, not the areas. The areas were 120 and 36 — those already got used up.

## ◆ Core

TWO CHUNKS EACH. WRITE BOTH BEFORE YOU ADD THEM.

1. 128 ÷ 4 .................... .................... = _______
2. 156 ÷ 6 .................... .................... = _______
3. 144 ÷ 3 .................... .................... = _______
4. 175 ÷ 5 .................... .................... = _______
5. 168 ÷ 7 .................... .................... = _______

## ★ Challenge

**C1.** A rectangle has area 252 and one side 7. Find the other side using two chunks.

....................

**C2.** Which is bigger, 144 ÷ 6 or 168 ÷ 8? Decide first, then check.

....................

**C3.** Divide 96 by 6 using only chunks of ten sixes. How many steps did it take? Now do it in one step.

NAME ___________

DATE ___________

# When It Doesn't Fit

The quotient and the remainder are two different answers. Read which one is being asked for.

## Warm-Up

REMAINDER ONLY — DON'T WRITE THE QUOTIENT

| | | | | | |
|---|---|---|---|---|---|
| 17 ÷ 5 | 23 ÷ 4 | 38 ÷ 6 | 50 ÷ 7 | 29 ÷ 3 | 45 ÷ 8 |
| ____ | ____ | ____ | ____ | ____ | ____ |

THE PIECE THAT WON'T FIT

94 ÷ 4 = 23 remainder 2

23 columns of 4 = 92 | r2

The dashed sliver is too short to make a full column. Draw it every time — don't just write “r2”.

## Core

BOTH ANSWERS EVERY TIME, THEN THE CHECK

1. 17 ÷ 5 q ______ r ______ check: 5 × ____ + ____ = 17
2. 94 ÷ 4 q ______ r ______ check: 4 × ____ + ____ = 94
3. 137 ÷ 6 q ______ r ______ check: 6 × ____ + ____ = 137
4. 83 ÷ 9 q ______ r ______ check: 9 × ____ + ____ = 83

## Challenge

**C1.** 26 kids, vans that hold 6. How many vans? What happened to the leftover?

**C2.** 26 cookies, 6 friends, whole cookies only. How many each? What happened to the leftover this time?

**C3.** Same numbers, different answers. Write the rule you'd give another kid for deciding which to do.

# Chunks Get Tidy

NAME ___________

DATE ___________

Three-digit dividends. Take the biggest chunk you can see, then finish what's left.

## Warm-Up FRIENDLY HUNDREDS

| | | | | | |
|---|---|---|---|---|---|
| 120 ÷ 4 | 150 ÷ 5 | 240 ÷ 6 | 350 ÷ 7 | 180 ÷ 9 | 160 ÷ 8 |
| ____ | ____ | ____ | ____ | ____ | ____ |

## Core SHOW EVERY CHUNK YOU TOOK. ESTIMATE FIRST.

1. 372 ÷ 3 = ________

2. 465 ÷ 5 = ________

3. 728 ÷ 8 = ________

4. 594 ÷ 6 = ________

5. 952 ÷ 7 = ________ Estimate before you start — roughly how big will the answer be? ..............................

## Challenge

**C1.** 845 ÷ 4 = ______ r ______. Show the chunks you used.

**C2.** Check your answer to 952 ÷ 7 by multiplying it back. Write the check as one equation.

**C3.** Invent a division with a three-digit dividend that comes out exactly, with no remainder. Prove it does.

NAME ___________

DATE ___________

# Remainder Race & Puzzles

No Warm-Up today. Play the game first, then take the puzzles as slowly as you like.

## Remainder Race · two players, one deck

Strip the face cards. Each player turns two cards to make a two-digit number, then rolls a die for the divisor. Divide. **Your score for the round is the remainder.** Ten rounds, highest total wins. A clean division scores zero — which means the safe-looking answer is the worst one.

After three rounds, stop and answer this: which divisor gives you the best chance of a big score, and why?

## ★ Puzzles

REASON IT OUT — NO GUESSING

P1 $100 \div 7 =$ _____ r _____

P2 $7 \times \square = 91$ $\square =$ _____

P3 $\square \div 6 = 14$ $\square =$ _____

P4 $9 \times \square = 108$ $\square =$ _____

P5 Smallest number over 50 that 7 divides exactly: ______

P6 $\square 6 \div 4 = 24$ $\square =$ _____

## Two decisions, one division

Both of these are 53 ÷ 8. The answers are different. Write each one and say why.

**a.** 53 people at a party, tables seat 8. How many tables?

**b.** You have $53 and tickets cost $8. How many tickets?

## The real question

Divide lots of different numbers by 5 and list every remainder you get. Which remainders are possible? Which are impossible? Explain why a remainder of 5 can never happen.

# The Rest of the Mission

Same three-tier shape as Week 1, every day. Objectives and problem sets outlined here; full worksheets follow the Week 1 pattern.

## Week 2 · Remainders

THE BIG QUESTION GETS ANSWERED THIS WEEK

**2.1 Mon**
Round up. Vans, boats, buses — when two leftovers force one more.

**2.2 Tue**
Drop it. Whole cookies each; the leftover stays on the plate.

**2.3 Wed**
Share it out. The remainder becomes a fraction of the thing.

**2.4 Thu**
The remainder is the answer: clocks, patterns, every-nth questions.

**Fri**
Snack Run planning. Real quantities, every decision written down.

## Week 3 · Long Division

MID-UNIT QUIZ FRIDAY · 85% TO ADVANCE

**3.1 Mon**
Under the bar. Partial quotients shrink into the standard layout.

**3.2 Tue**
Three-digit dividends. Estimate before you start, every time.

**3.3 Wed**
Zeros in the quotient. 618 ÷ 6 catches everybody once.

**3.4 Thu**
Check by multiplying back. Self-marking from here on.

**Fri**
Mid-unit quiz, 8 items, Weeks 1–3.

## Week 4 · Divisibility & Big Numbers

NEVER SKIP THE TWO-DIGIT DIVISORS

**4.1 Mon**
Tests for 2, 5, 10. Why the last digit is enough on its own.

**4.2 Tue**
Tests for 3 and 9 — and why adding the digits works at all.

**4.3 Wed**
Four-digit dividends. 3,472 ÷ 8 without flinching.

**4.4 Thu**
Two-digit divisors. Estimate, adjust, estimate again.

**Fri**
Perfect numbers: 6 = 1 + 2 + 3. Hunt the next one.

## Week 5 · Proof & Project

UNIT TEST FRIDAY · BADGE AWARDED

**5.1 Mon**
Missing-digit division. Reason backwards from the quotient.

**5.2 Tue**
Snack Run: split real quantities among real people.

**5.3 Wed**
Remainder defence. Justify every leftover call out loud.

**Thu**
Error journal sweep. Fix only what repeats.

**Fri**
Mission 02 test: 12 items + one explanation.

MATERIALS, WHOLE UNIT

Base-ten blocks · counters or dried beans · two dice · deck of cards · graph paper (¼ in) · the Snack Run shopping list · the error journal

WEEKLY CHECK

Five items every Friday, mixed tiers. Below 4/5 means Monday is a reteach, not new material.

THE MOST COMMON STALL

A right quotient with the wrong decision about the remainder. Ask “what were you dividing?” before you mark anything wrong.

# Mid-Unit Quiz

8 items · Weeks 1–3 · 7 of 8 keeps you flying

NAME ___________

DATE ___________

___ / 8

1. $63 \div 9 =$ ____________

**2.** Draw $84 \div 4$ as a rectangle. Label both chunks and write the answer.

**3.** $156 \div 6$. Any method you like — but show the method.

**4.** $94 \div 4 =$ ______ r ______. Then check it.

**5.** 29 tennis balls, tubes hold 4. How many tubes to store them all?

**6.** $372 \div 3 =$ ______

**7.** Which of 2, 3, 5, 9 divide 234? Name the test you used.

CHALLENGE ITEM 8. $\square 6 \div 4 = 24 \quad \square =$ _____

Clue you used first: ____________

NAME ___________

DATE ___________

# Mission 02 Test

12 items + one explanation · part 1 of 2

1. $72 \div 8 =$ ..............................

2. $54 \div 6 =$ ..............................

**3.** $96 \div 4$ with a rectangle. = ________

**4.** $175 \div 5$, show your chunks. = ________

**5.** $137 \div 6 =$ ______ r ______

**6.** $465 \div 5$, any method. = ________

**7.** 3,472 ÷ 8. Estimate first, then compute.

.............................. ..............................

**8.** 47 kids on a field trip, buses hold 9. How many buses?

..............................

..............................

**9.** You have $47. Comics cost $9. How many can you buy?

..............................

..............................

# Reasoning & the Big Question

ITEMS ___ / 12
EXPLANATION ___ / 4

**10.** Which of 2, 3, 5, 9, 10 divide 450? Show the test you used for each.

**11.** Someone says 728 ÷ 8 = 91. Are they right? Prove it without dividing again.

CHALLENGE ITEM **12.** 1☐5 ÷ 5 = 23 ☐ = _____ First clue:

EXPLAIN YOUR THINKING · WORTH 4 POINTS

## When you have a remainder, what do you do with it — and who decides?

Take 26 ÷ 6. Write two different real situations that give two different right answers, and say what made them different. Draw the leftover in at least one. Spelling doesn't count. Reasoning does.

TROPHY BAND · CIRCLE ONE

**85–100%**
Mission Commander
Leftover Badge earned. Mission 03 opens — and Challenge becomes your Core.

**70–84%**
Flight Engineer
Two targeted days on the flagged items, then retake those items only.

**Below 70%**
Ground Crew
Rebuild Week 1 with counters before returning to the long-division bar.

# Key A · Pre-Assessment, Lessons 1.1–1.2

## Pre-Assessment

A1 **7** A2 **7**

B1 rectangle 4 by 21, split 80 + 4 → **21**

B2 120 + 36 → 20 + 6 = **26**

C1 **23 r2**

C2 **5 vans** — 4 vans hold 24, two kids still need seats

D1 **124**

D2 **211 r1**

E1 **2, 3, 9** (not 5)

E2 **Yes,** 7 × 13 = 91

F1 **14 each, 2 left** — accept any sensible fate for the 2

F2 ▢ **= 9** (96 ÷ 4 = 24)

**Scoring note.** C2 and F1 are about the decision, not the arithmetic. A correct quotient with no reasoning scores half. Mark the strand partial and keep Week 2.

## Lesson 1.1 · Two Kinds of Division

**Warm-Up** 7 · 7 · 7 · 9 · 9 · 9

**Core**

1. 84 ÷ 4 = 20 + 1 = **21**
2. 96 ÷ 6 = 10 + 6 = **16**
3. 75 ÷ 5 = 10 + 5 = **15**
4. 90 ÷ 6 = 10 + 5 = **15**
5. 78 ÷ 3 = 20 + 6 = **26** (60 and 18)

**Lesson 1.1 Challenge.** C1 · 16 (96 ÷ 6; accept 60 ÷ 6 + 36 ÷ 6). C2 · 9 boxes; this is grouping — the group size is known and the number of groups is not. C3 · accept any two stories where one shares among a known number of people and the other makes groups of a known size, e.g. “48 stickers shared by 8 kids” versus “48 stickers into bags of 8.”

## Lesson 1.2 · The Missing Side

**Warm-Up** 8 · 9 · 9 · 4 · 7 · 12

**Core** (chunks · answer)

1. 120 ÷ 4 = 30, 8 ÷ 4 = 2 → **32**
2. 120 ÷ 6 = 20, 36 ÷ 6 = 6 → **26**
3. 120 ÷ 3 = 40, 24 ÷ 3 = 8 → **48**
4. 150 ÷ 5 = 30, 25 ÷ 5 = 5 → **35**
5. 140 ÷ 7 = 20, 28 ÷ 7 = 4 → **24**

**Challenge.** C1 · 36 (210 ÷ 7 = 30, then 42 ÷ 7 = 6). C2 · 144 ÷ 6 = 24 is bigger; 168 ÷ 8 = 21. Strong reasoning notices that 168 is only a little more than 144 while 8 is much more than 6. C3 · ten sixes four times gets to 240 — too far; the honest answer is one chunk of ten (60) then six more sixes, or in one step sixteen sixes. Accept any correct chunking; the point is that bigger chunks mean fewer steps.

**Watch for:** adding the areas instead of the widths — writing 156 ÷ 6 = 156. If that happens, go back to counters for one day.

# Key B · Lessons 1.3–1.4, Enrichment, Quiz

## Lesson 1.3 · When It Doesn't Fit

**Warm-Up** (remainders) 2 · 3 · 2 · 1 · 2 · 5

**Core** (quotient · remainder · check)

1. **3 r2** · 5 × 3 + 2 = 17
2. **23 r2** · 4 × 23 + 2 = 94
3. **22 r5** · 6 × 22 + 5 = 137
4. **9 r2** · 9 × 9 + 2 = 83

**Challenge.** C1 · 5 vans; the leftover forced an extra van. C2 · 4 cookies each; the leftover was dropped. C3 · accept any rule that names the situation as the decider — "if the leftover still needs looking after, round up; if it just sits there, drop it."

## Lesson 1.4 · Chunks Get Tidy

**Warm-Up** 30 · 30 · 40 · 50 · 20 · 20

1. 300÷3 + 60÷3 + 12÷3 = **124**
2. 450÷5 + 15÷5 = **93**
3. 720÷8 + 8÷8 = **91**
4. 540÷6 + 54÷6 = **99**
5. 700÷7 + 210÷7 + 42÷7 = **136**

**Challenge.** C1 · **211 r1** (800 ÷ 4 = 200, 44 ÷ 4 = 11, 1 left). C2 · 7 × 136 = 952, so the check closes exactly. C3 · any dividend built as divisor × whole number; the proof is the multiplication written back.

## Friday · Remainder Race

**Puzzles** P1 **14 r2** · P2 ▢ = **13** · P3 ▢ = **84**
P4 ▢ = **12** · P5 **56** · P6 ▢ = **9**

**Game question.** A divisor of 6 gives the best shot at a big score — the possible remainders run 0 to 5, so the ceiling is highest. Dividing by 2 can only ever score 0 or 1. Accept any answer that spots that the biggest possible remainder is always one less than the divisor.

**Two decisions.** a · **7 tables** (six seat 48, five people still standing). b · **6 tickets**, with $5 left over. Same division, opposite treatment of the leftover.

**The real question.** Dividing by 5, the only possible remainders are 0, 1, 2, 3, 4. A remainder of 5 is impossible because another whole 5 would fit — the quotient was simply too small. Accept any wording that reaches "the remainder is always smaller than the divisor."

## Mid-Unit Quiz · out of 8

1. **7**
2. 84 ÷ 4 = 20 + 1 = **21**
3. 156 ÷ 6 = 20 + 6 = **26**
4. **23 r2** · check 4 × 23 + 2 = 94
5. **8 tubes** (7 hold 28, one ball over)
6. **124**
7. **2, 3, 9** — even; digits sum to 9
8. ▢ = **9** (96 ÷ 4 = 24)

**7/8 or better → continue.** Below that, reteach only the items missed. A miss on 4 or 5 means remainders, not division — spend the two days on Week 2, not Week 1.

TEACHER PAGE · ANSWER KEY

# Key C · Mission 02 Test & Scoring

## Items 1–12 · one point each

1. **9**
2. **9**
3. 96 ÷ 4 = 20 + 4 = **24**
4. 175 ÷ 5 = 30 + 5 = **35**
5. **22 r5**
6. 465 ÷ 5 = 90 + 3 = **93**
7. estimate ≈ 400 · **434**
8. **6 buses** (five hold 45, two kids over)
9. **5 comics**, $2 left
10. **all five** — 450 is even, digits sum to 9, ends in 0
11. Yes — 8 × 91 = 728, remainder 0
12. ▢ = **1** (115 ÷ 5 = 23)

## Explanation · 4 points

Score the reasoning, not the writing. Award one point each:

- ☐ 1 · Computes 26 ÷ 6 = 4 remainder 2 correctly
- ☐ 2 · Gives two real situations with two different answers
- ☐ 3 · Says which one forces rounding up, and why
- ☐ 4 · States that the situation decides, not the arithmetic

**Model answer, kid-sized.** “The division always says 4 remainder 2. If they're vans I need a fifth van, because two kids can't walk. If they're cookies everybody gets 4 and two are spare. The numbers don't decide — what I'm dividing decides.”

**Total 16 points.** 14+ Mission Commander · 11–13 Flight Engineer · 10 or fewer Ground Crew. Item 12 missed alone never blocks the badge. Items 8 and 9 missed together means Week 2 gets revisited before Mission 03.

## Error Journal · photocopy this page

One row per mistake, for the whole mission. The “why” line must name a step. “I was careless” doesn't count.

| DATE | PROBLEM | I GOT / ANSWER | WHY IT HAPPENED — ONE SENTENCE |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |